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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 25th June Morning Shift

JEE Main Question Paper with Answer Key

25 Jun 2022	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	20 questions
Physics	27 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?

(A) NO

(B) N_2

(C) O_2

(D) C_2

(E) B_2

Choose the most appropriate answer from the options given below :

(A) (A), (B), (C) only

(B) (B), (C), (E) only

(C) (A), (C) only

(D) (D) only

Open solution

Q2. MCQ (Single Correct Answer)

The pair, in which ions are isoelectronic with Al^{3+} is :

(A) Br^- and Be^{2+}

(B) Cl^- and Li^+

(C) S^{2-} and K^+

(D) O^{2-} and Mg^{2+}

Open solution

Q3. MCQ (Single Correct Answer)

Number of electron deficient molecules among the following

PH_3, B_2H_6, CCl_4, NH_3, LiH and BCl_3 is

(A) 0

(B) 1

(C) 2

(D) 3

Open solution

Q4. MCQ (Single Correct Answer)

White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of :

(A) $[Ag(NH_3)_4]Cl_2$

(B) $[Ag(Cl)_2(NH_3)_2]$

(C) $[Ag(NH_3)_2]Cl$

(D) $[Ag(NH_3)Cl]Cl$

Open solution

Q5. MCQ (Single Correct Answer)

Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?

(A) It will not prefer to undergo redox reactions.

(B) It will prefer to gain electron and act as an oxidizing agent.

(C) It will prefer to give away an electron and behave as reducing agent.

(D) It acts as both, oxidizing and reducing agent.

Open solution

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Q6. MCQ (Single Correct Answer)

Among the following, which is the strongest oxidizing agent?

(A) Mn^{3+} (B) Fe^{3+} (C) Ti^{3+} (D) Cr^{3+}

Open solution

Q7. MCQ (Single Correct Answer)

Phenol on reaction with dilute nitric acid, gives two products. Which method will be most efficient for large scale separation?

(A) Chromatographic separation
(B) Fractional Crystallisation
(C) Steam distillation
(D) Sublimation

Open solution

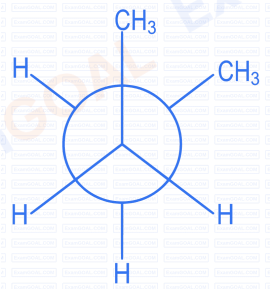
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Q8. MCQ (Single Correct Answer)

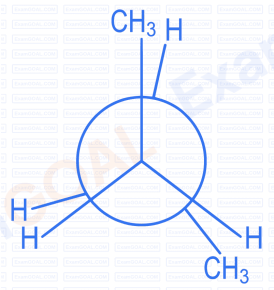
Open solution

In the following structures, which one is having staggered conformation with maximum dihedral angle?

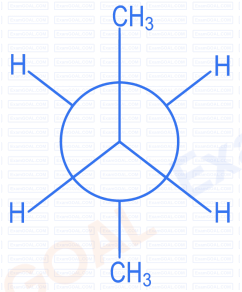
(A)



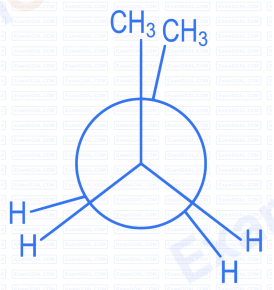
(B)



(C)



(D)

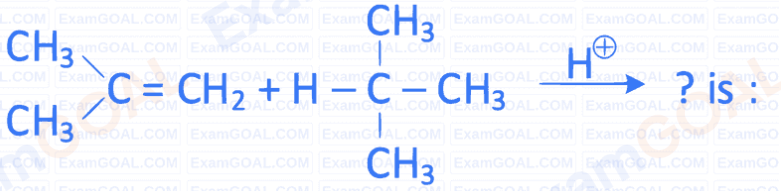


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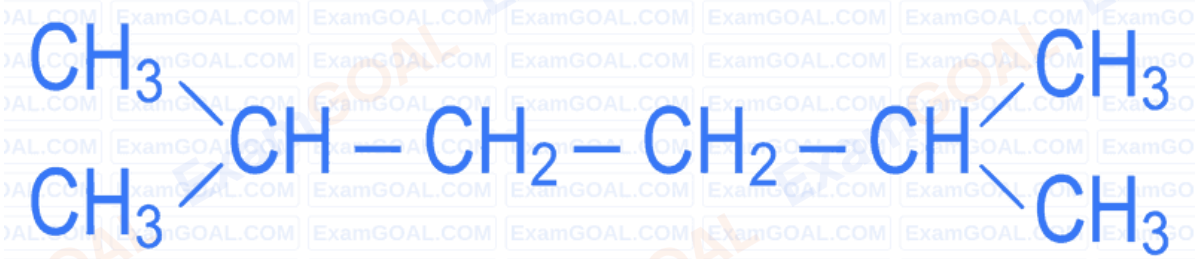
Q9. MCQ (Single Correct Answer)

Open solution

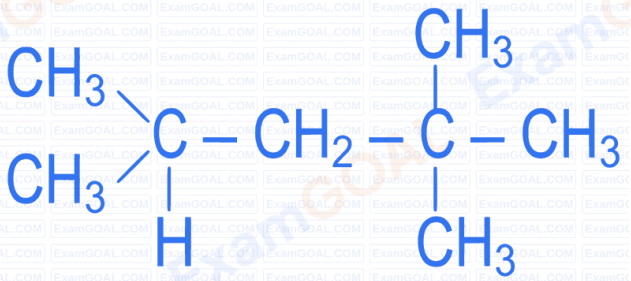
The product formed in the following reaction.



(A)



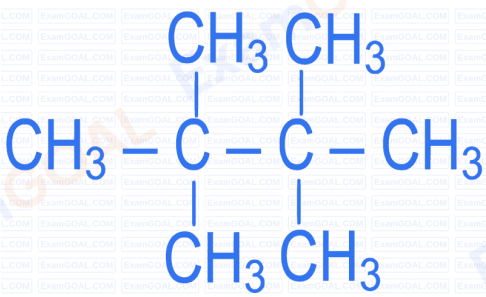
(B)



(C)



(D)



Q10. MCQ (Single Correct Answer)

Open solution

The IUPAC name of ethylidene chloride is :

(A) 1-Chloroethene

(B) 1-Chloroethyne

(C) 1,2-Dichloroethane

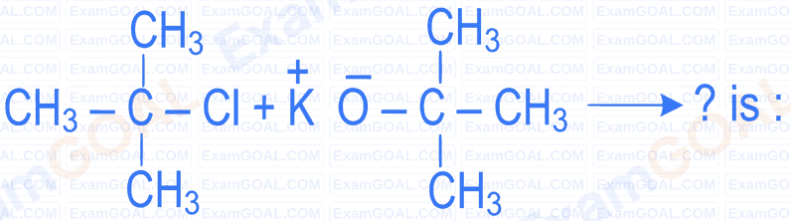
(D) 1,1-Dichloroethane

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Q11. MCQ (Single Correct Answer)

Open solution

The major product in the reaction



(A) t-Butyl ethyl ether

(B) 2,2-Dimethyl butane

(C) 2-Methyl pent-1-ene

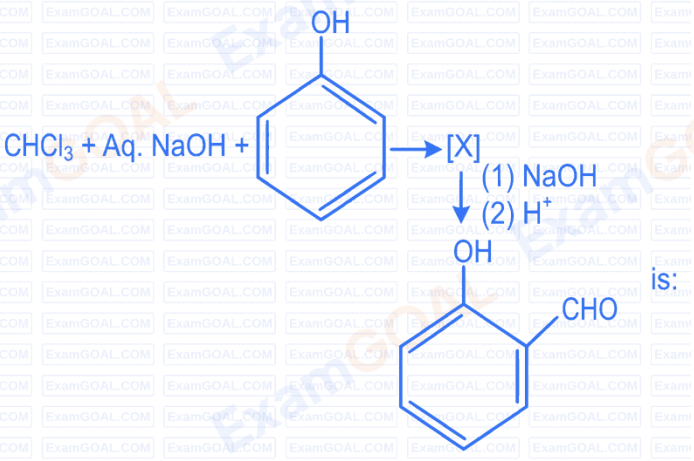
(D) 2-Methyl prop-1-ene

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Q12. MCQ (Single Correct Answer)

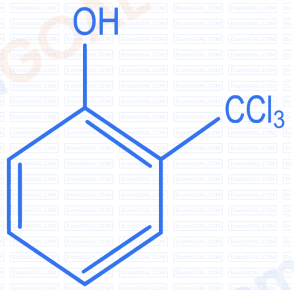
Open solution

The intermediate X, in the reaction :

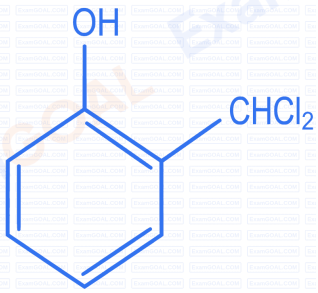


is:

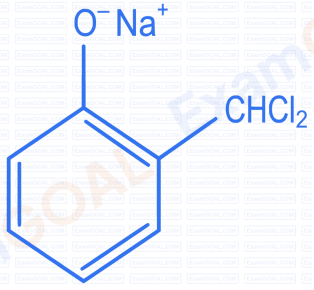
(A)



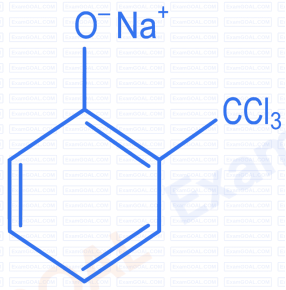
(B)



(C)



(D)

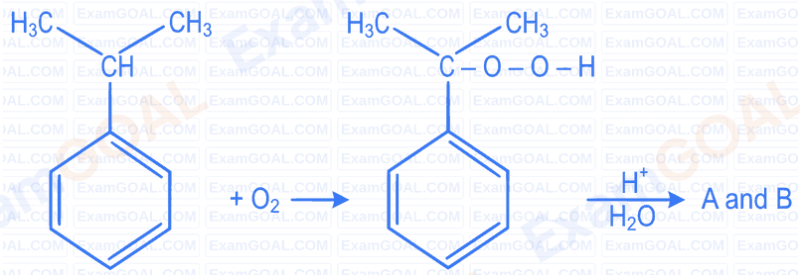


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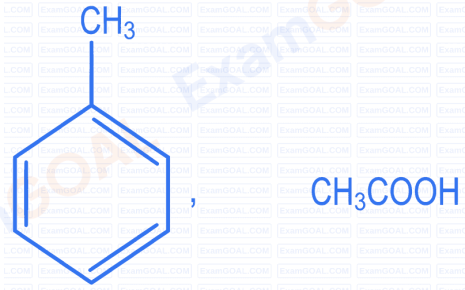
Q13. MCQ (Single Correct Answer)

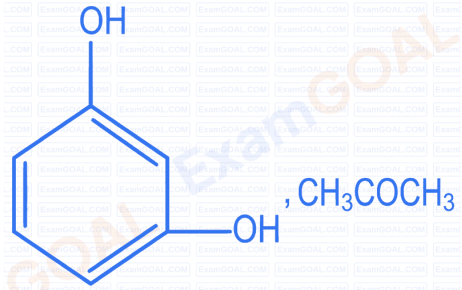
Open solution

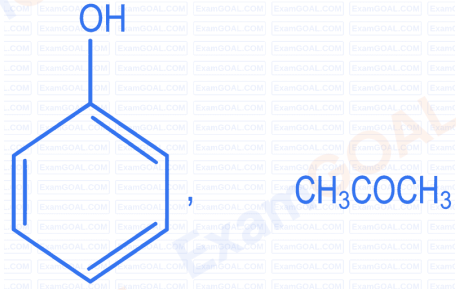
In the following reaction :

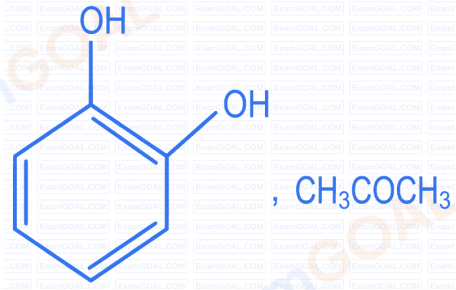


The compounds A and B respectively are :

(A)  , CH_3COOH

(B)  , CH_3COCH_3

(C)  , CH_3COCH_3

(D)  , CH_3COCH_3

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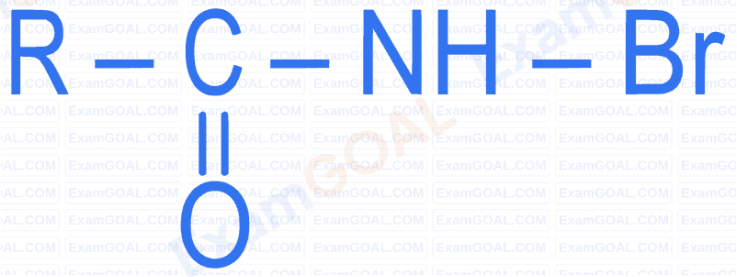
Q14. MCQ (Single Correct Answer)

Open solution

The reaction of



with bromine and KOH gives RNH_2 as the end product. Which one of the following is the intermediate product formed in this reaction?

(A) 

(B) R – NH – Br

(C) R – N = C = O

(D) 

Q15. Numerical

Open solution

The number of N atoms in 681 g of $C_7H_5N_3O_6$ is $x \times 10^{21}$. The value of x is ($N_A = 6.02 \times 10^{23}$ mol^{-1}) (Nearest Integer)

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Q16. Numerical

The longest wavelength of light that can be used for the ionisation of lithium atom (Li) in its ground state is x $\times 10^{-8}$ m. The value of x is _____. (Nearest Integer).

(Given : Energy of the electron in the first shell of the hydrogen atom is -2.2×10^{-18} J ; h = 6.63 $\times$ 10^{-34} Js and c = 3 $\times 10^{8}$ ms^{-1})

Open solution

Q17. Numerical

The standard entropy change for the reaction

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is -550 J K^{-1} at 298 K.

[Given : The standard enthalpy change for the reaction is -165 kJ mol^{-1}]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer)

Open solution

Q18. Numerical

1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is _____ $\times 10^3$ m mols.

Open solution

Q19. Numerical

The standard free energy change (ΔG°) for 50% dissociation of N_2O_4 into NO_2 at 27°C and 1 atm pressure is $-$ x J mol^{-1}. The value of x is _____. (Nearest Integer)

[Given : R = 8.31 J K^{-1} mol^{-1}, log 1.33 = 0.1239, ln 10 = 2.3]

Open solution

Q20. Numerical

In a cell, the following reactions take place

$Fe^{2+} \rightarrow Fe^{3+} + e^- \quad E^o_{Fe^{3+}/Fe^{2+}} = 0.77\,V$

$2I^- \rightarrow I_2 + 2e^- \quad E^o_{I_2/I^-} = 0.54\,V$

The standard electrode potential for the spontaneous reaction in the cell is x $\times 10^{-2}$ V at 298 K. The value of x is _____. (Nearest Integer)

Open solution

Q21. Numerical

For a given chemical reaction

$\gamma_1A + \gamma_2B \rightarrow \gamma_3C + \gamma_4D$

Concentration of C changes from 10 mmol dm^{-3} to 20 mmol dm^{-3} in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be 9 mmol dm^{-3} s^{-1}. Therefore, the rate of reaction is _____ mmol dm^{-3} s^{-1}. (Nearest Integer)

Open solution

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Q22. Numerical

If $[Cu(H_2O)_4]^{2+}$ absorbs a light of wavelength 600 nm for d-d transition, then the value of octahedral crystal field splitting energy for $[Cu(H_2O)_6]^{2+}$ will be _____ $\times 10^{-21}$ J. [Nearest Integer]

(Given : h = 6.63 $\times$ 10^{-34} Js and c = 3.08 $\times$ 10^{8} ms^{-1})

Open solution

Q23. Numerical

Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) [Nearest Integer]

Open solution

Mathematics 20 questions

Q24. MCQ (Single Correct Answer)

The value of $\int_0^{\pi} \frac{e^{\cos x}\sin x}{(1+\cos^2 x)(e^{\cos x}+e^{-\cos x})}dx$ is equal to:

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Open solution

Q25. MCQ (Single Correct Answer)

Let f : N $\rightarrow$ R be a function such that $f(x+y) = 2f(x)f(y)$ for natural numbers x and y. If f(1) = 2, then the value of α for which

$\sum_{k=1}^{10} f(\alpha+k) = \frac{512}{3}(2^{20}-1)$

holds, is :

(A) 2 (B) 3 (C) 4 (D) 6

Open solution

Q26. MCQ (Single Correct Answer)

Let A be a 3 $\times$ 3 real matrix such that

$A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}; A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix}$ and $A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}$.

If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A-2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$ has :

(A) no solution (B) infinitely many solutions

(C) unique solution (D) exactly two solutions

Open solution

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Q27. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as $f(x) = x^3 + x - 5$. If g(x) is a function such that $f(g(x)) = x, \forall' x' \in R$, then g'(63) is equal to _____.

(A) $\frac{1}{49}$ (B) $\frac{3}{49}$ (C) $\frac{43}{49}$ (D) $\frac{91}{49}$

Open solution

Q28. MCQ (Single Correct Answer)

If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \ldots\ldots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$, then the remainder when K is divided by 6 is :

(A) 1 (B) 2 (C) 3 (D) 5

Open solution

Q29. MCQ (Single Correct Answer)

Let f(x) be a polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value of $\lim\limits_{x \to 1} \frac{f(x)}{x-1}$ is equal to:

(A) -15 (B) -60 (C) 60 (D) 15

Open solution

Q30. MCQ (Single Correct Answer)

Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1|E_2) = \frac{1}{2}$, $P(E_2|E_1) = \frac{3}{4}$ and $P(E_1 \cap E_2) = \frac{1}{8}$. Then :

(A) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ (B) $P(E_1' \cap E_2') = P(E_1') \cdot P(E_2)$

(C) $P(E_1 \cap E_2') = P(E_1) \cdot P(E_2)$ (D) $P(E_1' \cap E_2) = P(E_1) \cdot P(E_2)$

Open solution

Q31. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M = \sum\limits_{k=1}^{10} A^{2k}$ and $N = \sum\limits_{k=1}^{10} A^{2k-1}$ then MN^2 is :

(A) a non-identity symmetric matrix

(B) a skew-symmetric matrix

(C) neither symmetric nor skew-symmetric matrix

(D) an identity matrix

Open solution

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Q32. MCQ (Single Correct Answer)

Let $g : (0, \infty) \to R$ be a differentiable function such that

$\int \left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)(e^x + 1 - xe^x)}{(e^x + 1)^2} \right) dx = \frac{x\, g(x)}{e^x + 1} + c$, for all x > 0, where c is an arbitrary constant. Then :

Open solution

(A) g is decreasing in $\left(0, \frac{\pi}{4}\right)$

(B) g' is increasing in $\left(0, \frac{\pi}{4}\right)$

(C) g + g' is increasing in $\left(0, \frac{\pi}{2}\right)$

(D) g − g' is increasing in $\left(0, \frac{\pi}{2}\right)$

Q33. MCQ (Single Correct Answer)

Let $f : R \to R$ and $g : R \to R$ be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1 - 2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha - 1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

Open solution

(A) (2, 3) (B) $(-2, -1)$ (C) (1, 2) (D) $(-1, 1)$

Q34. MCQ (Single Correct Answer)

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0$, $i = 1, 2, 3$ be a vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a}, \vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to:

Open solution

(A) $\sqrt{7}$ (B) $\sqrt{2}$ (C) 2 (D) 7

Q35. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation $(x + 1)y' - y = e^{3x}(x + 1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve $y = y(x)$ is :

Open solution

(A) not a critical point (B) a point of local minima

(C) a point of local maxima (D) a point of inflection

Q36. MCQ (Single Correct Answer)

If the solution curve $y = y(x)$ of the differential equation $y^2dx + (x^2 - xy + y^2)dy = 0$, which passes through the point (1, 1) and intersects the line $y = \sqrt{3}x$ at the point $(\alpha, \sqrt{3}\alpha)$, then value of $\log_e(\sqrt{3}\alpha)$ is equal to :

Open solution

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

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Q37. MCQ (Single Correct Answer)

Let $x = 2t$, $y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim\limits_{t \to 1} k$ is equal to :

Open solution

(A) $\frac{17}{18}$ (B) $\frac{19}{18}$ (C) $\frac{11}{18}$ (D) $\frac{13}{18}$

Q38. MCQ (Single Correct Answer)

Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then $arg(z)$ is equal to :

Open solution

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$ (B) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$

(C) $\tan^{-1}(3) - \pi$ (D) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

Q39. Numerical

The number of 3-digit odd numbers, whose sum of digits is a multiple of 7, is _____.

Open solution

Q40. Numerical

Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $\left|\left(\vec{a} - \vec{b}\right) \times \left(\vec{a} + \vec{b}\right)\right|^2 + 4\left(\vec{a} . \vec{b}\right)^2$ is equal to _____.

Open solution

Q41. Numerical

Let the abscissae of the two points P and Q be the roots of $2x^2 - rx + p = 0$ and the ordinates of P and Q be the roots of $x^2 - sx - q = 0$. If the equation of the circle described on PQ as diameter is $2(x^2 + y^2) - 11x - 14y - 22 = 0$, then $2r + s - 2q + p$ is equal to _____.

Open solution

Q42. Numerical

Let $f : R \to R$ be a function defined by

$f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to g(1) is _____.

Open solution

Q43. Numerical

Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____.

Open solution

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Physics 27 questions

Q44. MCQ (Single Correct Answer)

If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be :

(A) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

(B) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

(C) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$

(D) $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Open solution

Q45. MCQ (Single Correct Answer)

$\overrightarrow{A}$ is a vector quantity such that $|\overrightarrow{A}|$ = non-zero constant. Which of the following expression is true for $\overrightarrow{A}$?

(A) $\overrightarrow{A}.\overrightarrow{A} = 0$

(B) $\overrightarrow{A} \times \overrightarrow{A} < 0$

(C) $\overrightarrow{A} \times \overrightarrow{A} = 0$

(D) $\overrightarrow{A} \times \overrightarrow{A} > 0$

Open solution

Q46. MCQ (Single Correct Answer)

Which of the following relations is true for two unit vectors $\widehat{A}$ and $\widehat{B}$ making an angle θ to each other?

(A) $|\widehat{A} + \widehat{B}| = |\widehat{A} - \widehat{B}| \tan\frac{\theta}{2}$

(B) $|\widehat{A} - \widehat{B}| = |\widehat{A} + \widehat{B}| \tan\frac{\theta}{2}$

(C) $|\widehat{A} + \widehat{B}| = |\widehat{A} - \widehat{B}| \cos\frac{\theta}{2}$

(D) $|\widehat{A} - \widehat{B}| = |\widehat{A} + \widehat{B}| \cos\frac{\theta}{2}$

Open solution

Q47. MCQ (Single Correct Answer)

If force $\overrightarrow{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be :

(A) $3\hat{i} + 4\hat{j} - 2\hat{k}$

(B) $-10\hat{i} + 10\hat{j} + 5\hat{k}$

(C) $10\hat{i} + 5\hat{j} - 10\hat{k}$

(D) $10\hat{i} + \hat{j} - 5\hat{k}$

Open solution

Q48. MCQ (Single Correct Answer)

The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be : (Given g = acceleration due to gravity at the surface of earth).

(A) g/2 (B) g/4 (C) g/3 (D) g/9

Open solution

Q49. MCQ (Single Correct Answer)

The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as :

(A) $r^{1/2}$ (B) r (C) r^2 (D) r^3

Open solution

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Q50. MCQ (Single Correct Answer)

Open solution

The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be :

(A) $v_{rms} = \sqrt{\frac{2}{3}} v_p$

(B) $v_{rms} = \sqrt{\frac{3}{2}} v_p$

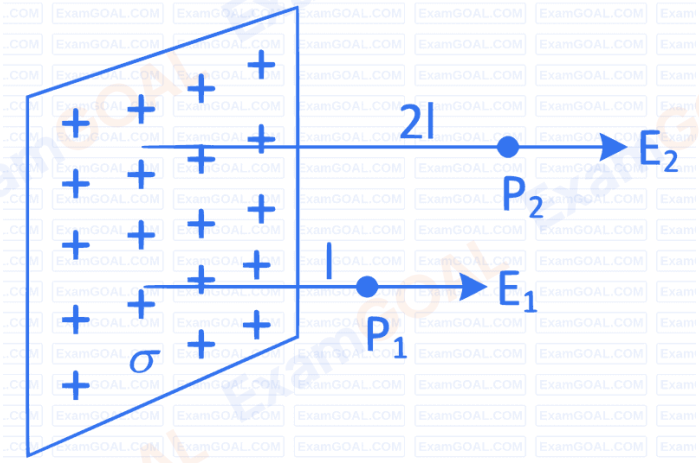
(C) $v_{rms} = v_p$

(D) $v_{rms} = \sqrt{\frac{1}{3}} v_p$

Q51. MCQ (Single Correct Answer)

Open solution

In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and 2l from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



(A) $E_1 = \sigma/\varepsilon_0,\ E_2 = \sigma/2\varepsilon_0$

(B) $E_1 = 2\sigma/\varepsilon_0,\ E_2 = \sigma/\varepsilon_0$

(C) $E_1 = E_2 = \sigma/2\varepsilon_0$

(D) $E_1 = E_2 = \sigma/\varepsilon_0$

Q52. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II.

	List - I		List -II
(A)	AC generator	(I)	Detects the presence of current in the circuit
(B)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(C)	Transformer	(III)	Works on the principle of resonance in AC circuit
(D)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the correct answer from the options given below :

(A) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(B) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

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Q53. MCQ (Single Correct Answer)

A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r (r < R) from its centre will be :

(A) $B \propto r^2$ (B) $B \propto r$ (C) $B \propto \frac{1}{r^2}$ (D) $B \propto \frac{1}{r}$

Open solution

Q54. MCQ (Single Correct Answer)

If wattless current flows in the AC circuit, then the circuit is :

(A) Purely Resistive circuit (B) Purely Inductive circuit

(C) LCR series circuit (D) RC series circuit only

Open solution

Q55. MCQ (Single Correct Answer)

The electric field in an electromagnetic wave is given by $E = 56.5 \sin \omega(t - x/c)\ NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space.

(Given : $\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

(A) $5.65\ Wm^{-2}$ (B) $4.24\ Wm^{-2}$

(C) $1.9 \times 10^{-7}\ Wm^{-2}$ (D) $56.5\ Wm^{-2}$

Open solution

Q56. MCQ (Single Correct Answer)

The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

(A) 2 I (B) 6 I (C) 5 I (D) 7 I

Open solution

Q57. MCQ (Single Correct Answer)

A light wave travelling linearly in a medium of dielectric constant 4, is incident on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :

(Given : relative permeability of medium $\mu_r = 1$)

(A) 10° (B) 20° (C) 30° (D) 60°

Open solution

Q58. MCQ (Single Correct Answer)

The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be :

(A) 1 : 1 (B) 1 : 2 (C) 4 : 1 (D) 2 : 1

Open solution

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Q59. MCQ (Single Correct Answer)

The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because :

(A) fractional change in majority carriers produce higher forward bias current

(B) fractional change in majority carriers produce higher reverse bias current

(C) fractional change in minority carriers produce higher forward bias current

(D) fractional change in minority carriers produce higher reverse bias current

Open solution

Q60. MCQ (Single Correct Answer)

The difference of speed of light in the two media A and B ($v_A - v_B$) is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum c = 3×10^8 ms^{-1})

(A) 1.303 (B) 1.318 (C) 1.13 (D) 0.12

Open solution

Q61. MCQ (Single Correct Answer)

A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G?

(A) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

(B) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistances.

(C) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistances.

(D) $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistances.

Open solution

Q62. Numerical

A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _____ m.

Open solution

Q63. Numerical

A 0.5 kg block moving at a speed of 12 ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be _____ Nm^{-1}.

Open solution

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Q64. Numerical

The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is _____ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)

Open solution

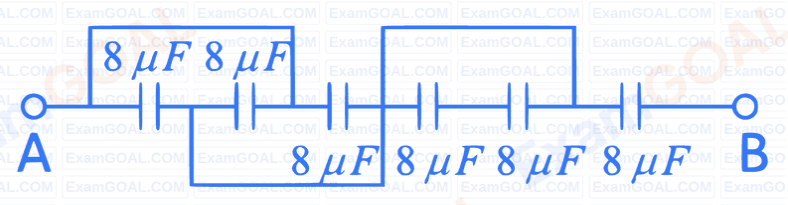
Q65. Numerical

The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is _____ cm.

Open solution

Q66. Numerical

The equivalent capacitance between points A and B in below shown figure will be _____ μF.



Open solution

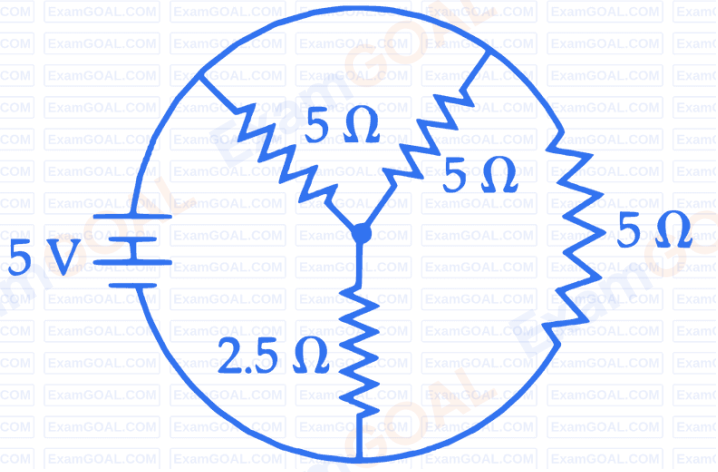
Q67. Numerical

A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J.

Open solution

Q68. Numerical

The total current supplied to the circuit as shown in figure by the 5 V battery is _____ A.



Open solution

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Q69. Numerical

The current in a coil of self inductance 2.0 H is increasing according to I = 2 sin(t^2) A. The amount of energy spent during the period when current changes from 0 to 2 A is _____ J.

Open solution

Q70. Numerical

A force on an object of mass 100 g is $\left(10\hat{i}+5\hat{j}\right)$ N. The position of that object at t = 2 s is $\left(a\hat{i}+b\hat{j}\right)$ m after starting from rest. The value of $\frac{a}{b}$ will be _____.

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Incorrect statement for Tyndall effect is :

(A) The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude.

(B) The diameter of the dispersed particles is much smaller than the wavelength of the light used.

(C) During projection of movies in the cinemas hall, Tyndall effect is noticed.

(D) It is used to distinguish a true solution from a colloidal solution.

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Leaching of gold with dilute aqueous solution of NaCN in presence of oxygen gives complex [A], which on reaction with zinc forms the elemental gold and another complex [B]. [A] and [B], respectively are :

(A) $[Au(CN)_4]^-$ and $[Zn(CN)_2(OH)_2]^{2-}$

(B) $[Au(CN)_2]^-$ and $[Zn(OH)_4]^{2-}$

(C) $[Au(CN)_2]^-$ and $[Zn(CN)_4]^{2-}$

(D) $[Au(CN)_4]^{2-}$ and $[Zn(CN)_6]^{4-}$

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following alkaline earth metal ions has the highest ionic mobility in its aqueous solution?

(A) Be^{2+} (B) Mg^{2+} (C) Ca^{2+} (D) Sr^{2+}

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

The eutrophication of water body results in :

(A) loss of Biodiversity.

(B) breakdown of organic matter.

(C) increase in biodiversity.

(D) decrease in BOD.

Open solution

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes, because :

(A) soap particles remain floating in water as ions.

(B) the hydrophobic part of soap is not able to take away grease.

(C) the micelles are not formed due to concentration of soap, below its CMC value.

(D) colloidal structure of soap in water is completely disturbed.

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following is an example of artificial sweetner?

(A) Bithional (B) Alitame (C) Salvarsan (D) Lactose

Open solution

Q77. Numerical · **Chemistry**

The distance between Na^+ and Cl^- ions in solid NaCl of density 43.1 g cm^{-3} is _____ $\times 10^{-10}$ m. (Nearest Integer)

(Given : $N_A = 6.02 \times 10^{23}$ mol^{-1})

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let a circle C touch the lines $L_1 : 4x - 3y + K_1 = 0$ and $L_2 : 4x - 3y + K_2 = 0, K_1, K_2 \in R$. If a line passing through the centre of the circle C intersects L_1 at $(-1, 2)$ and L_2 at $(3, -6)$, then the equation of the circle C is :

(A) $(x-1)^2 + (y-2)^2 = 4$ (B) $(x+1)^2 + (y-2)^2 = 4$

(C) $(x-1)^2 + (y+2)^2 = 16$ (D) $(x-1)^2 + (y-2)^2 = 16$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let a, b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$. If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to :

(A) $\frac{5}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) 1

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

Consider the following two propositions:

$P1 :\sim (p \rightarrow \sim q)$

$P2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then :

(A) P1 is TRUE and P2 is FALSE

(B) P1 is FALSE and P2 is TRUE

(C) Both P1 and P2 are FALSE

(D) Both P1 and P2 are TRUE

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If $y = m_1x + c_1$ and $y = m_2x + c_2, m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8|m_1m_2|$ is equal to :

(A) $3 + 4\sqrt{2}$ (B) $-5 + 6\sqrt{2}$ (C) $-4 + 3\sqrt{2}$ (D) $7 + 6\sqrt{2}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : x + y + z = 5. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

(A) 2 (B) 5 (C) 7 (D) 11

Open solution

Q83. Numerical · **Mathematics**

Let C_r denote the binomial coefficient of x^r in the expansion of $(1 + x)^{10}$. If for $\alpha, \beta \in R$,

$C_1 + 3.2C_2 + 5.3C_3 +$ ..... upto 10 terms $= \frac{\alpha \times 2^{11}}{2^\beta - 1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots\ldots \; upto\, 10\, terms\right)$ then the value of $\alpha + \beta$ is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which

$14\cos ec^2x - 2\sin^2x = 21 - 4\cos^2x$ holds, is _____.

Open solution

Q85. Numerical · **Mathematics**

For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is _____.

Open solution

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Q86. Numerical · **Mathematics**

Open solution

Let the lines

$L_1 : \vec{r} = \lambda\left(\hat{i} + 2\hat{j} + 3\hat{k}\right),\ \lambda \in R$

$L_2 : \vec{r} = \left(\hat{i} + 3\hat{j} + \hat{k}\right) + \mu\left(\hat{i} + \hat{j} + 5\hat{k}\right);\ \mu \in R,$

intersect at the point S. If a plane ax + by − z + d = 0 passes through S and is parallel to both the lines L_1 and L_2, then the value of a + b + d is equal to _____.

Q87. Numerical · **Mathematics**

Open solution

The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}$, ... is equal to _____.

Q88. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements :

Statement I : Davisson-Germer experiment establishes the wave nature of electrons.

Statement II : If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the correct answer from the option given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

A signal of 100 THz frequency can be transmitted with maximum efficiency by :

(A) Coaxial cable

(B) Optical fibre

(C) Twisted pair of copper wires

(D) Water

Q90. Numerical · **Physics**

Open solution

A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is 540 cal g^{-1}, then the heat rejected by the steam engine per minute is _____ $\times 10^3$ cal.

(Given : specific heat capacity of water : 1 cal g^{-1} $°C^{-1}$)

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Answer Key

Chemistry

1. **C**	2. **D**	3. **C**	4. **C**	5. **B**
6. **A**	7. **C**	8. **C**	9. **B**	10. **D**
11. **D**	12. **C**	13. **C**	14. **C**	15. **5418**
16. **4**	17. **300**	18. **0**	19. **710**	20. **23**
21. **1**	22. **766**	23. **1143**		

Mathematics

24. **C**	25. **C**	26. **B**	27. **A**	28. **D**
29. **A**	30. **C**	31. **A**	32. **D**	33. **A**
34. **B**	35. **B**	36. **C**	37. **D**	38. **B**
39. **63**	40. **576**	41. **7**	42. **2**	43. **414**

Physics

44. **C**	45. **C**	46. **B**	47. **B**	48. **D**
49. **C**	50. **B**	51. **C**	52. **A**	53. **B**
54. **B**	55. **B**	56. **B**	57. **D**	58. **D**
59. **D**	60. **C**	61. **B**	62. **2**	63. **600**
64. **100**	65. **80**	66. **6**	67. **450**	68. **2**
69. **4**	70. **2**			

Out of syllabus

71. **B**	72. **C**	73. **D**	74. **A**	75. **C**
76. **B**	77. **1**	78. **C**	79. **A**	80. **C**
81. **C**	82. **B**	83. **286**	84. **4**	85. **4**
86. **5**	87. **98**	88. **A**	89. **B**	90. **31**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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